Estimated Annual Agricultural Pesticide Use
Oregon, Statewide Ranking 2014-2018

21 July 2026

Summary

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides **statewide in Oregon** during 2018, as reported by USGS Estimated Annual Agricultural Pesticide Use data, summed across all Oregon counties. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification in `data/pesticide_classification.csv`, shared unchanged with the Washington pesticide class trends chapter). Rankings and estimates are based on the *EPest-high method* described by Thelin and Stone (2013) and Baker and Stone (2015):

> "EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD." *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. "detailed interpretation of where and how much use occurs within a county is not appropriate"); summing county estimates to a statewide total inherits this caveat.
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 included once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
ALS	ALS-Inhibitor (Sulfonylurea & Related)
AVM	Avermectin
BIO	Biological/Microbial
BIP	Bipyridylium
CB	Carbamate
CAA	Chloroacetanilide
DMI	DMI Fungicide (Triazole/Imidazole)
DIA	Diamide
DNA	Dinitroaniline
DTC	Dithiocarbamate
FUM	Fumigant
GLY	Glycine/Phosphonate
INO	Inorganic/Mineral
IGR	Insect Growth Regulator

Table 1: Chemical class abbreviation key (used in the Class column of Table 2) *(continued)*

Code	Chemical class
NEO	Neonicotinoid
OC	Organochlorine
OP	Organophosphate
FUN	Other Fungicide
HRB	Other Herbicide
INS	Other Insecticide/Miticide
OTH	Other/Unclassified
PHX	Phenoxy
PHP	Phenylpyrazole
PGR	Plant Growth Regulator
PYR	Pyrethroid
SDH	SDHI
SPN	Spinosyn
QOI	Strobilurin (QoI)
TC	Thiocarbamate
TRZ	Triazine

Table 2: Top 80 EPest-high Estimates Statewide (OR), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank (OR)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM	FUM	1848035	2106124	1957167
2	DICHLOROPROPENE	FUM	1043404	942486	1086891
3	CHLOROPICRIN	FUM	978674	383420	256838
4	BACILLUS AMYLOLIQUEFACIEN	BIO	941297	445492	445492
5	PETROLEUM OIL	INS	604008	726426	795341
6	GLYPHOSATE	GLY	522544	556282	555285
7	METAM POTASSIUM	FUM	480182	615752	633563
8	SULFUR	INO	357260	313104	340140
9	SULFURIC ACID	INO	203322	443563	439492
10	PARAQUAT	BIP	141380	92816	69335
11	CALCIUM POLYSULFIDE	INO	109533	122287	108542
12	2,4-D	PHX	105047	143330	148567
13	MANCOZEB	DTC	100922	124134	128417
14	ETHOPROPHOS	OP	97921	71030	47113
15	TRIFLURALIN	DNA	72524	56736	79358
16	BURKHOLDERIA SPP	BIO	64767	79674	53260
17	MCPA	PHX	56977	76353	63885
18	FLUFENACET	CAA	55004	38710	35878
19	ATRAZINE	TRZ	50786	63960	54356
20	COPPER	INO	47360	34752	33510
21	EPTC	TC	44020	60405	74153
22	CHLOROTHALONIL	FUN	42797	68114	57907
23	GLUFOSINATE	GLY	41551	27926	29435
24	PENDIMETHALIN	DNA	36739	35917	37232
25	KAOLIN CLAY	INO	36524	312823	272474
26	METOLACHLOR & METOLACHLOR-S	CAA	36204	35139	40537
27	BROMOXYNIL	HRB	35039	29268	29658
28	METOLACHLOR-S	CAA	34903	32989	37243
29	DIURON	HRB	33981	72426	79627
30	COPPER SULFATE	INO	32547	45155	41049
31	CHLORPYRIFOS	OP	31918	38050	39659
32	ZIRAM	DTC	31415	12762	12647
33	METRIBUZIN	TRZ	30696	22205	26280
34	AZOXYSTROBIN	QOI	30196	19566	16937
35	DICAMBA	HRB	28555	23140	22080

Table 2: Top 80 EPest-high Estimates Statewide (OR), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank (OR)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
36	METHOMYL	CB	27114	28565	28125
37	PROPICONAZOLE	DMI	26282	19731	17272
38	DCPA	HRB	22237	30674	32472
39	OXAMYL	CB	16394	9948	22507
40	COPPER HYDROXIDE	INO	15064	20119	18665
41	THIOPHANATE-METHYL	FUN	13951	18987	17471
42	CLOPYRALID	HRB	13927	11020	11727
43	DIMETHENAMID & DIMETHENAMID-P	CAA	13278	17194	15293
44	DIMETHENAMID-P	CAA	13278	13615	13145
45	MALATHION	OP	12602	15752	19509
46	COPPER OXYCHLORIDE	INO	11509	7658	5972
47	DIMETHOATE	OP	11470	18621	17955
48	PYRACLOSTROBIN	QOI	9248	9280	9896
49	SIMAZINE	TRZ	8934	6782	10143
50	TRICLOPYR	HRB	8208	10744	11495
51	DIQUAT	BIP	8104	5444	4212
52	FLUROXYPYR	HRB	8066	13511	11058
53	CAPTAN	FUN	7743	15295	22238
54	BOSCALID	SDH	7279	6215	7401
55	ETHOFUMESATE	HRB	6735	10641	8341
56	CYHALOTHRIN-LAMBDA	PYR	6440	4684	3966
57	PINOXADEN	HRB	6154	3176	2928
58	TERBACIL	HRB	6151	3077	3000
59	CLETHODIM	HRB	5586	6163	5476
60	BENTAZONE	HRB	5278	5033	6404
61	SAFLUFENACIL	HRB	5224	2080	1583
62	TRIFLUMIZOLE	DMI	5012	4822	5655
63	FLUOPYRAM	SDH	4653	2428	1809
64	PYMETROZINE	INS	4494	2416	1901
65	FLUTOLANIL	SDH	4489	3608	3583
66	BUPROFEZIN	IGR	4024	3433	3583
67	IMIDACLOPRID	NEO	3992	3737	4504
68	IPRODIONE	FUN	3936	7816	8932
69	PYROXASULFONE	HRB	3935	3212	3042
70	FLUMIOXAZIN	HRB	3836	2183	1982
71	MCPB	PHX	3768	1471	1231
72	BENZOVINDIFLUPYR	SDH	3722	1368	1368
73	ALPHA CYPERMETHRIN	PYR	3679	1921	1921
74	SPIROTETRAMAT	INS	3638	3940	3858
75	PYRIMETHANIL	FUN	3625	2858	2617
76	MEFENOXAM	FUN	3552	4376	3751
77	PYRIDABEN	INS	3493	4461	4091
78	PENTHIOPYRAD	SDH	3245	4339	4498
79	BACILLUS THURINGIENSIS	BIO	3126	2766	2229
80	FLUAZINAM	FUN	3103	1953	2300

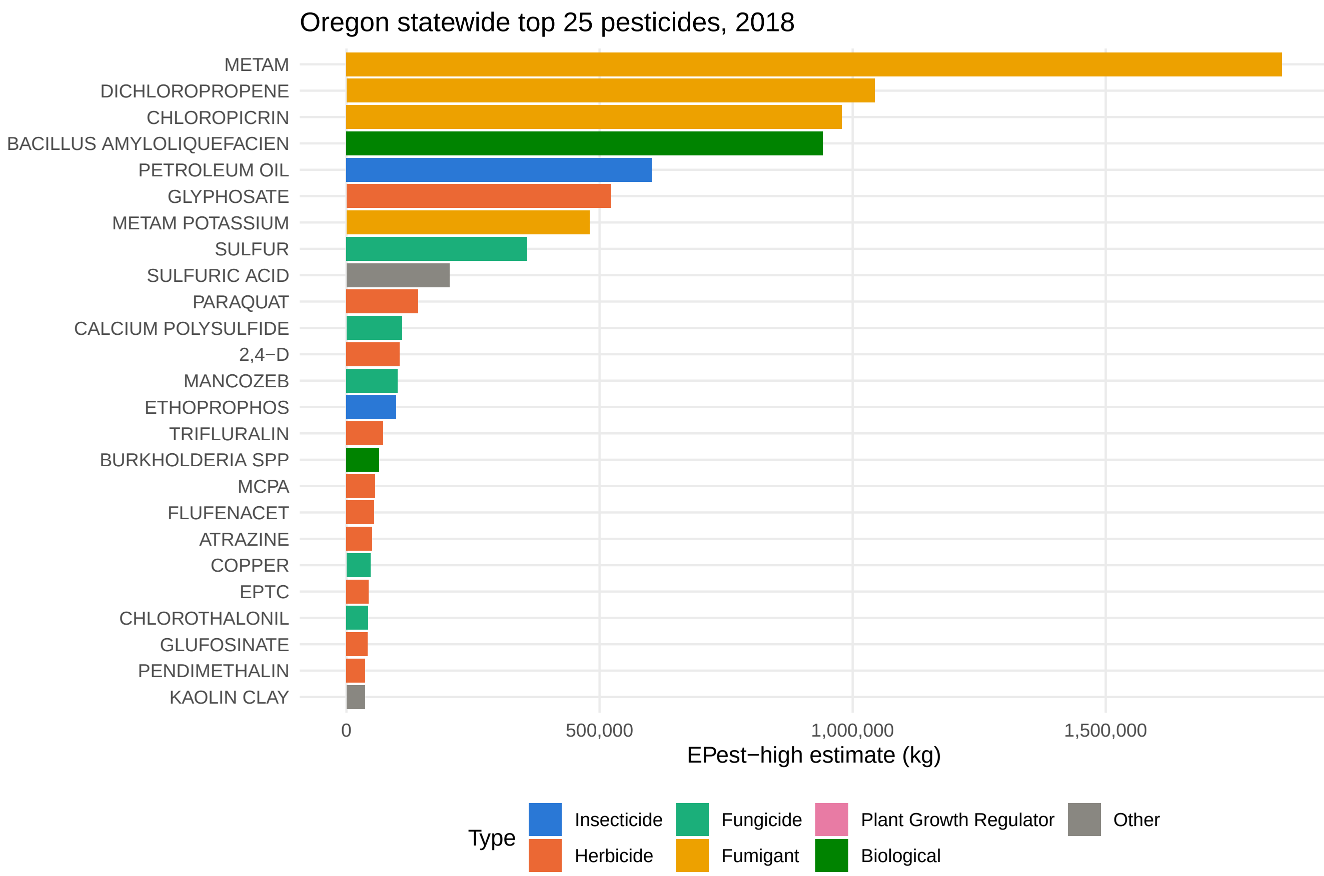

Figure 1: Top 25 compounds statewide by EPest-high mass applied, most recent year, colored by pesticide type.

Appendices

Table 3: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	1366649	1848035
2	DICHLOROPROPENE	1035770	1043404
3	CHLOROPICRIN	625811	978674
4	BACILLUS AMYLOLIQUEFACIEN	411266	941297
5	PETROLEUM OIL	584108	604008
6	GLYPHOSATE	498892	522544
7	METAM POTASSIUM	2491	480182
8	SULFUR	349892	357260
9	SULFURIC ACID	0	203322
10	PARAQUAT	75228	141380
11	CALCIUM POLYSULFIDE	107620	109533
12	2,4-D	97434	105047
13	MANCOZEB	93122	100922
14	ETHOPROPHOS	36647	97921
15	TRIFLURALIN	3430	72524
16	BURKHOLDERIA SPP	28320	64767
17	MCPA	29670	56977
18	FLUFENACET	14954	55004
19	ATRAZINE	29936	50786
20	COPPER	46663	47360
21	EPTC	37975	44020
22	CHLOROTHALONIL	38734	42797
23	GLUFOSINATE	41371	41551
24	PENDIMETHALIN	28738	36739
25	KAOLIN CLAY	36359	36524
26	METOLACHLOR & METOLACHLOR-S	33013	36204
27	BROMOXYNIL	32312	35039
28	METOLACHLOR-S	31712	34903
29	DIURON	15896	33981
30	COPPER SULFATE	32250	32547
31	CHLORPYRIFOS	25921	31918
32	ZIRAM	31415	31415
33	METRIBUZIN	20750	30696
34	AZOXYSTROBIN	12711	30196
35	DICAMBA	17076	28555
36	METHOMYL	20434	27114
37	PROPICONAZOLE	25374	26282
38	DCPA	4512	22237
39	OXAMYL	12815	16394
40	COPPER HYDROXIDE	13065	15064
41	THIOPHANATE-METHYL	8656	13951
42	CLOPYRALID	2876	13927
43	DIMETHENAMID & DIMETHENAMID-P	11252	13278
44	DIMETHENAMID-P	11252	13278
45	MALATHION	5754	12602
46	COPPER OXYCHLORIDE	9727	11509
47	DIMETHOATE	5076	11470
48	PYRACLOSTROBIN	3305	9248
49	SIMAZINE	8714	8934
50	TRICLOPYR	452	8208

Table 4: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	1617276	2106124
2	DICHLOROPROPENE	896139	942486
3	PETROLEUM OIL	719686	726426
4	METAM POTASSIUM	110013	615752
5	GLYPHOSATE	523365	556282
6	BACILLUS AMYLOLIQUEFACIEN	184003	445492
7	SULFURIC ACID	0	443563
8	CHLOROPICRIN	227162	383420
9	SULFUR	308810	313104
10	KAOLIN CLAY	122231	312823
11	2,4-D	122710	143330
12	MANCOZEB	121465	124134
13	CALCIUM POLYSULFIDE	119193	122287
14	PARAQUAT	57449	92816
15	BURKHOLDERIA SPP	34821	79674
16	MCPA	21583	76353
17	DIURON	23819	72426
18	ETHOPROPHOS	25975	71030
19	CHLOROTHALONIL	56620	68114
20	ATRAZINE	32039	63960
21	EPTC	47917	60405
22	TRIFLURALIN	2928	56736
23	COPPER SULFATE	43834	45155
24	FLUFENACET	10244	38710
25	CHLORPYRIFOS	31394	38050
26	PENDIMETHALIN	28446	35917
27	METOLACHLOR & METOLACHLOR-S	31622	35139
28	COPPER	33344	34752
29	PROPAMOCARB HCL	0	34495
30	METOLACHLOR-S	29472	32989
31	DCPA	18938	30674
32	BROMOXYNIL	23556	29268
33	METHOMYL	23721	28565
34	GLUFOSINATE	26332	27926
35	PHOSPHORIC ACID	8333	24917
36	DICAMBA	14484	23140
37	METRIBUZIN	16547	22205
38	COPPER HYDROXIDE	15806	20119
39	PROPICONAZOLE	18572	19731
40	AZOXYSTROBIN	9030	19566
41	THIOPHANATE-METHYL	10084	18987
42	DIMETHOATE	3372	18621
43	DIMETHENAMID & DIMETHENAMID-P	14084	17194
44	MALATHION	8419	15752
45	CAPTAN	10491	15295
46	DIMETHENAMID-P	10505	13615
47	FLUROXYPYR	2578	13511
48	ZIRAM	11868	12762
49	CLOPYRALID	1445	11020
50	QUINTOZENE	1551	10970

Table 5: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	1571319	1957167
2	DICHLOROPROPENE	953697	1086891
3	PETROLEUM OIL	789659	795341
4	METAM POTASSIUM	249285	633563
5	GLYPHOSATE	523355	555285
6	BACILLUS AMYLOLIQUEFACIEN	184003	445492
7	SULFURIC ACID	28778	439492
8	SULFUR	331445	340140
9	KAOLIN CLAY	152363	272474
10	CHLOROPICRIN	160941	256838
11	2,4-D	129887	148567
12	MANCOZEB	126600	128417
13	CALCIUM POLYSULFIDE	102382	108542
14	DIURON	27620	79627
15	TRIFLURALIN	2548	79358
16	EPTC	60300	74153
17	PARAQUAT	45113	69335
18	MCPA	22618	63885
19	CHLOROTHALONIL	47605	57907
20	ATRAZINE	28138	54356
21	BURKHOLDERIA SPP	23225	53260
22	ETHOPROPHOS	19654	47113
23	COPPER SULFATE	39453	41049
24	METOLACHLOR & METOLACHLOR-S	34664	40537
25	CHLORPYRIFOS	30953	39659
26	METOLACHLOR-S	31417	37243
27	PENDIMETHALIN	29595	37232
28	PHOSPHORIC ACID	23326	36309
29	FLUFENACET	7858	35878
30	PROPAMOCARB HCL	0	34495
31	COPPER	29282	33510
32	DCPA	16718	32472
33	BROMOXYNIL	21961	29658
34	GLUFOSINATE	20788	29435
35	METHOMYL	23081	28125
36	METRIBUZIN	17486	26280
37	OXAMYL	19788	22507
38	CAPTAN	14723	22238
39	DICAMBA	14744	22080
40	MALATHION	9977	19509
41	COPPER HYDROXIDE	14485	18665
42	DIMETHOATE	4403	17955
43	THIOPHANATE-METHYL	10822	17471
44	PROPICONAZOLE	16071	17272
45	AZOXYSTROBIN	9499	16937
46	METHYL BROMIDE	15680	15891
47	DIMETHENAMID & DIMETHENAMID-P	12279	15293
48	QUINTOZENE	6353	14054
49	DIMETHENAMID-P	10131	13145
50	ZIRAM	8276	12647